Computação Científica para a Indústria · IFSC

Mosquitos pelo som

Redes neurais profundas
para classificação acústica



Câmpus Florianópolis · foto institucional

Luiz Eduardo do Nascimento Ciarallo

INSTITUTO FEDERAL
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O voo deixa uma assinatura sonora

As asas produzem
padrões periódicos.

Objetivo: classificar
20 espécies pelo áudio.



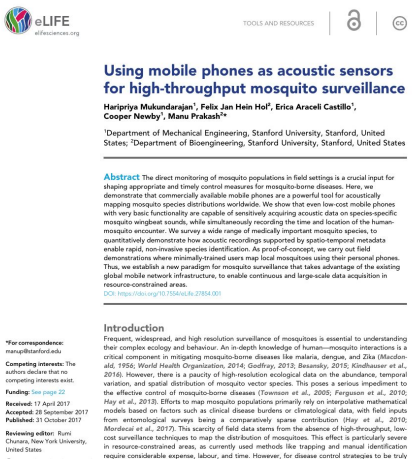
Aedes aegypti · fotografia CDC

Do som do voo à classificação por rede neural.

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Três estudos orientaram a proposta

Mukundarajan · 2017



eLIFE

TOOLS AND RESOURCES

Using mobile phones as acoustic sensors for high-throughput mosquito surveillance

Haripriya Mukundarajan[1], Felix Jan Hein Hol[2], Erica Araceli Castillo[1], Cooper Newby[1], Manu Prakash[2]*

[1]Department of Mechanical Engineering, Stanford University, Stanford, United States; [2]Department of Bioengineering, Stanford University, Stanford, United States

Abstract The direct monitoring of mosquito populations in field settings is a crucial input for shaping appropriate and timely control measures for mosquito-borne diseases. Here, we demonstrate that commercially available mobile phones are a powerful tool for acoustically mapping mosquito species distributions worldwide. We show that even low-cost mobile phones with very basic functionality are capable of sensitively acquiring acoustic data on species-specific mosquito wingbeat sounds, while simultaneously recording the time and location of the human-mosquito encounter. We survey a wide range of medically important mosquito species, to quantitatively demonstrate how acoustic recordings supported by spatio-temporal metadata enable rapid, non-invasive species identification. As proof-of-concept, we carry out field demonstrations where minimally-trained users map local mosquitoes using their personal phones. Thus, we establish a new paradigm for mosquito surveillance that takes advantage of the existing global mobile network infrastructure, to enable continuous and large-scale data acquisition in resource-constrained areas.

DOI: https://doi.org/10.7554/eLife.27854.001

Introduction

Frequent, widespread, and high resolution surveillance of mosquitoes is essential to understanding their complex ecology and behaviour. An in-depth knowledge of human—mosquito interactions is a critical component in mitigating mosquito-borne diseases like malaria, dengue, and Zika (*Macdonald, 1956*; *World Health Organization, 2014*; *Godfray, 2013*; *Besansky, 2015*; *Kindhauser et al., 2016*). However, there is a paucity of high-resolution ecological data on the abundance, temporal variation, and spatial distribution of mosquito vector species. This poses a serious impediment to the effective control of mosquito-borne diseases (*Townson et al., 2005*; *Ferguson et al., 2010*; *Hay et al., 2013*). Efforts to map mosquito populations primarily rely on interpolative mathematical models based on factors such as clinical disease burdens or climatological data, with field inputs from entomological surveys being a comparatively sparse contribution (*Hay et al., 2010*; *Mordecai et al., 2017*). This scarcity of field data stems from the absence of high-throughput, low-cost surveillance techniques to map the distribution of mosquitoes. This effect is particularly severe in resource-constrained areas, as currently used methods like trapping and manual identification require considerable expense, labour, and time. However, for disease control strategies to be truly

*For correspondence: manu@stanford.edu

Competing interests: The authors declare that no competing interests exist.

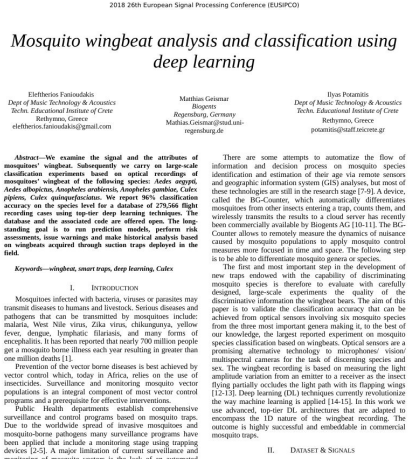
Funding: See page 22

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Reviewing editor: Rumi Chunara, New York University, United States

Captura com celulares

Fanioudakis · 2018



2018 26th European Signal Processing Conference (EUSIPCO)

Mosquito wingbeat analysis and classification using deep learning

Eleftherios Fanioudakis
Dept of Music Technology & Acoustics Techn. Educational Institute of Crete
Rethymno, Greece
eleftherios.fanioudakis@gmail.com

Matthias Geismar
Biogents
Regensburg, Germany
Matthias.Geismar@stud.uni-regensburg.de

Ilyas Potamitis
Dept of Music Technology & Acoustics Techn. Educational Institute of Crete
Rethymno, Greece
potamitis@staff.teicrete.gr

Abstract—We examine the signal and the attributes of mosquitoes' wingbeat. Subsequently we carry on large-scale classification experiments based on optical recordings of mosquitoes' wingbeat of the following species: *Aedes aegypti, Aedes albopictus, Anopheles arabiensis, Anopheles gambiae, Culex pipiens, Culex quinquefasciatus.* We report 96% classification accuracy on the species level for a database of 279,566 flight recording cases using top-tier deep learning techniques. The database and the associated code are offered open. The long-standing goal is to run prediction models, perform risk assessments, issue warnings and make historical analysis based on wingbeats acquired through suction traps deployed in the field.

Keywords—wingbeat, smart traps, deep learning, Culex

I. Introduction

Mosquitoes infected with bacteria, viruses or parasites may transmit diseases to humans and livestock. Serious diseases and pathogens that can be transmitted by mosquitoes include: malaria, West Nile virus, Zika virus, chikungunya, yellow fever, dengue, lymphatic filariasis, and many forms of encephalitis. It has been reported that nearly 700 million people get a mosquito borne illness each year resulting in greater than one million deaths [1].

Prevention of the vector borne diseases is best achieved by vector control which, today in Africa, relies on the use of insecticides. Surveillance and monitoring mosquito vector populations is an integral component of most vector control programs and a prerequisite for effective interventions.

Public Health departments establish comprehensive surveillance and control programs based on mosquito traps. Due to the worldwide spread of invasive mosquitoes and mosquito-borne pathogens many surveillance programs have been applied that include a monitoring stage using trapping devices [2-5]. A major limitation of current surveillance and

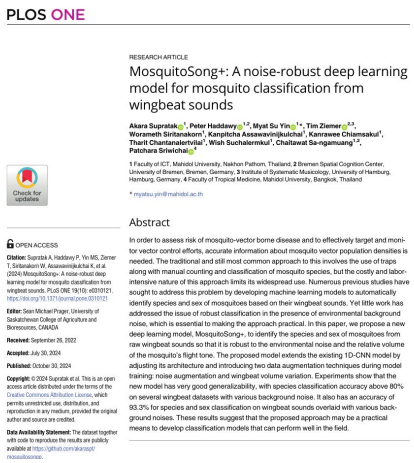
There are some attempts to automatize the flow of information and decision process on mosquito species identification and estimation of their age via remote sensors and geographic information system (GIS) analyses, but most of these technologies are still in the research stage [7-9]. A device, called the BG-Counter, which automatically differentiates mosquitoes from other insects entering a trap, counts them, and wirelessly transmits the results to a cloud server has recently been commercially available by Biogents AG [10-11]. The BG-Counter allows to remotely measure the dynamics of nuisance caused by mosquito populations to apply mosquito control measures more focused in time and space. The following step is to be able to differentiate mosquito genera or species.

The first and most important step in the development of new traps endowed with the capability of discriminating mosquito species is therefore to evaluate with carefully designed, large-scale experiments the quality of the discriminative information the wingbeat bears. The aim of this paper is to validate the classification accuracy that can be achieved from optical sensors involving six mosquito species from the three most important genera making it, to the best of our knowledge, the largest reported experiment on mosquito species classification based on wingbeats. Optical sensors are a promising alternative technology to microphones' vision/multispectral cameras for the task of discerning species and sex. The wingbeat recording is based on measuring the light amplitude variation from an emitter to a receiver as the insect flying partially occludes the light path with its flapping wings [12-13]. Deep learning (DL) techniques currently revolutionize the way machine learning is applied [14-15]. In this work we use advanced, top-tier DL architectures that are adapted to encompass the 1D nature of the wingbeat recording. The outcome is highly successful and embeddable in commercial mosquito traps.

II. Dataset & Signals

Redes profundas

Supratak · 2024



PLOS ONE

RESEARCH ARTICLE

MosquitoSong+: A noise-robust deep learning model for mosquito classification from wingbeat sounds

Akara Supratak[1], Peter Haddawy[1,2], Myat Su Yin[1]*, Tim Ziemer[3,4], Worameth Siritanakorn[1], Kanpitcha Assawavinijkulchai[1], Kanrawee Chiamsakul[1], Thanit Chantanalerdvitat[1], Wish Suchalermkul[1], Chaitawat Sa-ngamuang[1,2], Patchara Sriwichai[5]

1 Faculty of ICT, Mahidol University, Nakhon Pathom, Thailand, **2** Bremen Spatial Cognition Center, University of Bremen, Bremen, Germany, **3** Institute of Systematic Musicology, University of Hamburg, Hamburg, Germany, **4** Faculty of Tropical Medicine, Mahidol University, Bangkok, Thailand

* myatsu.yin@mahidol.ac.th

Abstract

In order to assess risk of mosquito-vector borne disease and to effectively target and monitor vector control efforts, accurate information about mosquito vector population densities is needed. The traditional and still most common approach to this involves the use of traps along with manual counting and classification of mosquito species, but the costly and labor-intensive nature of this approach limits its widespread use. Numerous previous studies have sought to address this problem by developing machine learning models to automatically identify species and sex of mosquitoes based on their wingbeat sounds. Yet little work has addressed the issue of robust classification in the presence of environmental background noise, which is essential to making the approach practical. In this paper, we propose a new deep learning model, MosquitoSong+, to identify the species and sex of mosquitoes from raw wingbeat sounds so that it is robust to the environmental noise and the relative volume of the mosquito's flight tone. The proposed model extends the existing 1D-CNN model by adjusting its architecture and introducing two data augmentation techniques during model training: noise augmentation and wingbeat volume variation. Experiments show that the new model has very good generalizability, with species classification accuracy above 80% on several wingbeat datasets with various background noise. It also has an accuracy of 93.3% for species and sex classification on wingbeat sounds overlaid with various background noises. These results suggest that the proposed approach may be a practical means to develop classification models that can perform well in the field.

Check for updates

OPEN ACCESS

Citation: Supratak A, Haddawy P, Yin MS, Ziemer T, Siritanakorn W, Assawavinijkulchai K, et al. (2024) MosquitoSong+: A noise-robust deep learning model for mosquito classification from wingbeat sounds. PLoS ONE 19(10): e0310121. https://doi.org/10.1371/journal.pone.0310121

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Data Availability Statement: The dataset together with code to reproduce the results are publicly available at https://github.com/akarasupt/mosquitosongs.

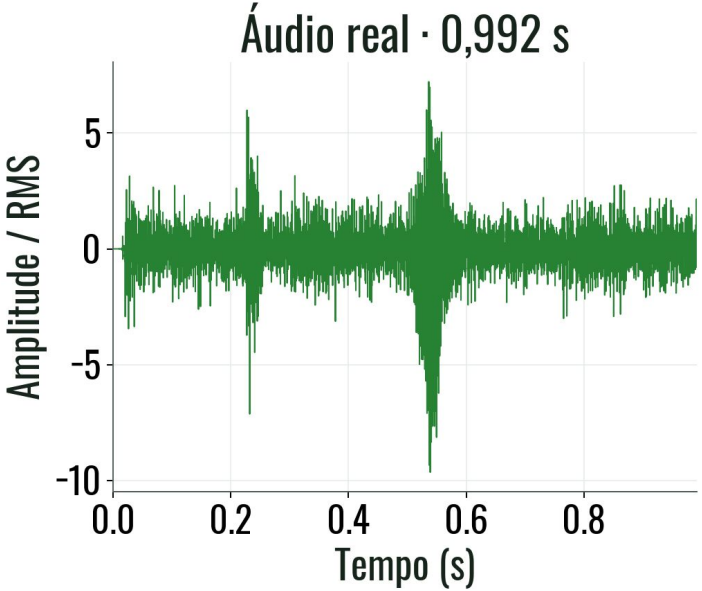
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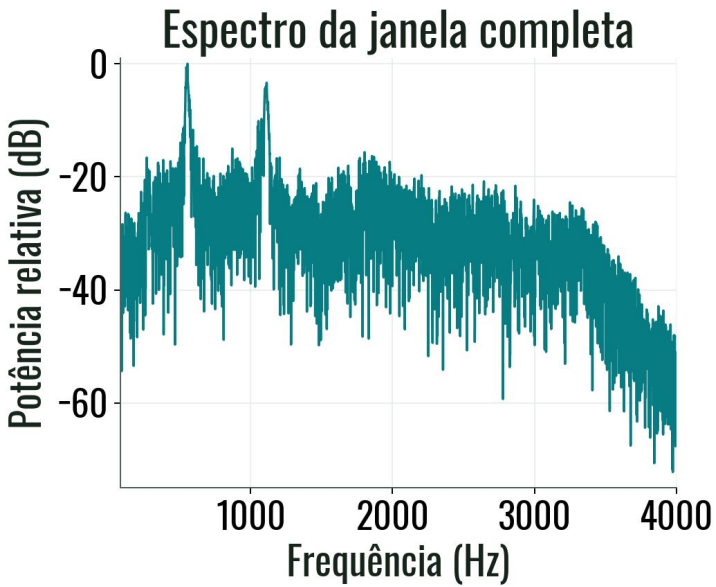
MosquitoSong+ · ruído

Literatura: captura acústica, redes neurais e robustez ao ruído.

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Primeiro, observar o áudio e o espectro

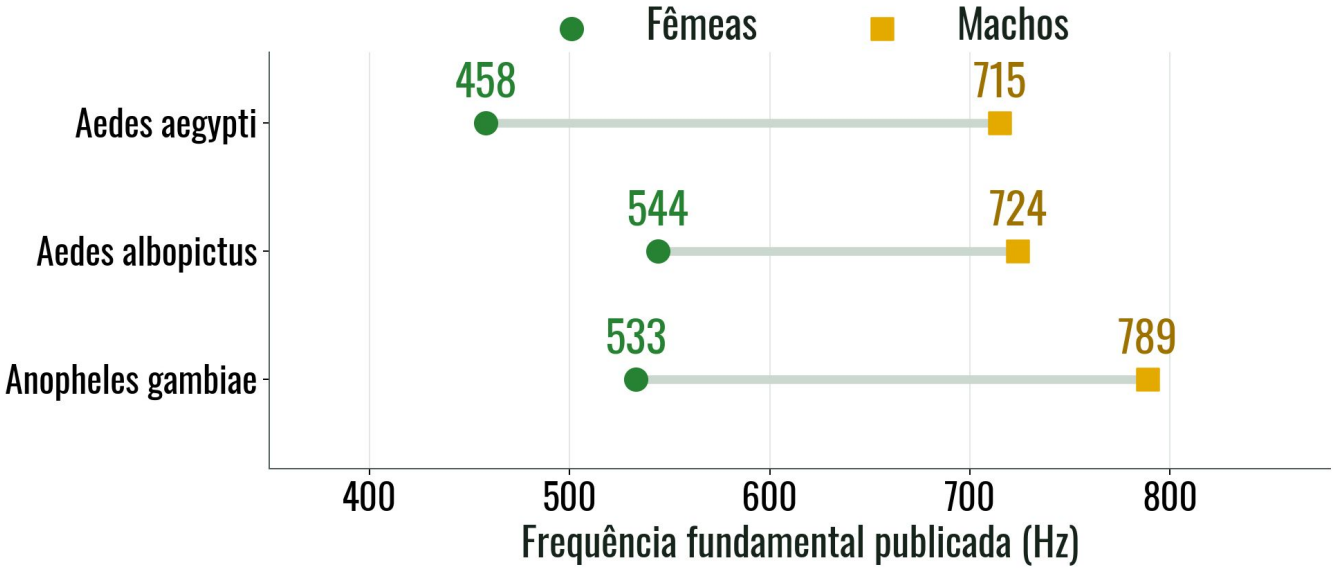




Aedes aegypti · áudio real Dryad · 16 kHz · 15.872 amostras por janela.

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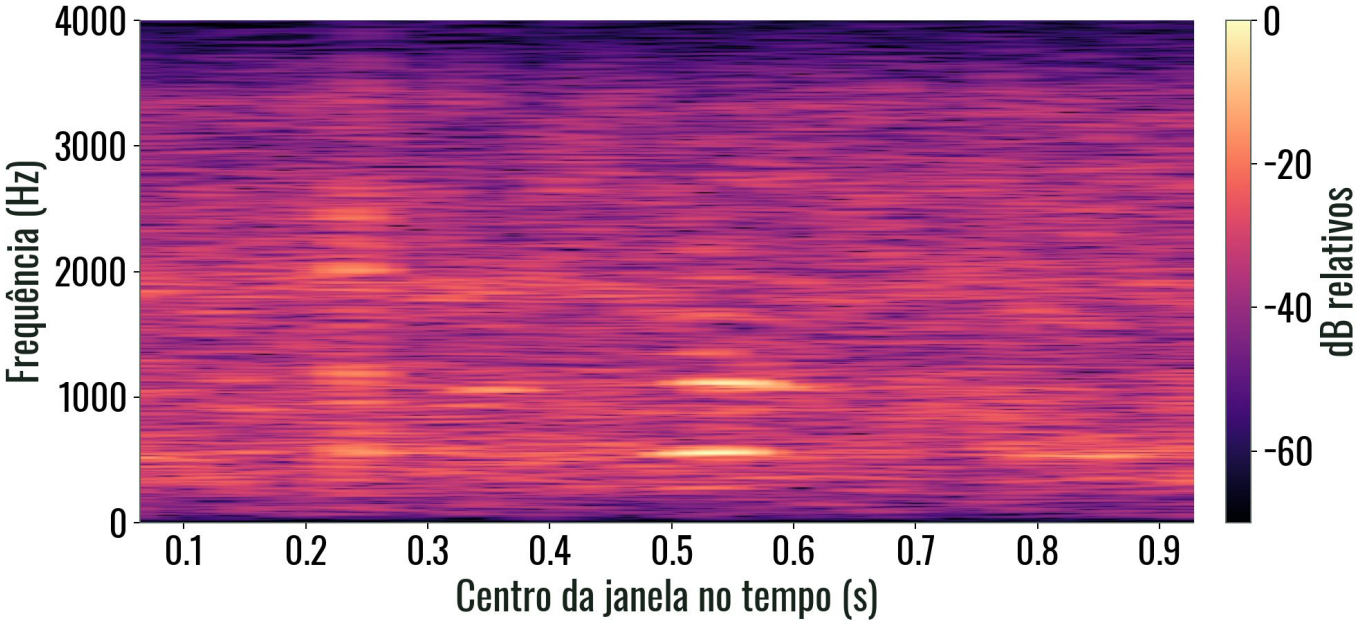
A frequência ajuda a descrever o voo



Observações a 25 °C · Sinka et al. (2021), Tabela 1 · valores em Hz.

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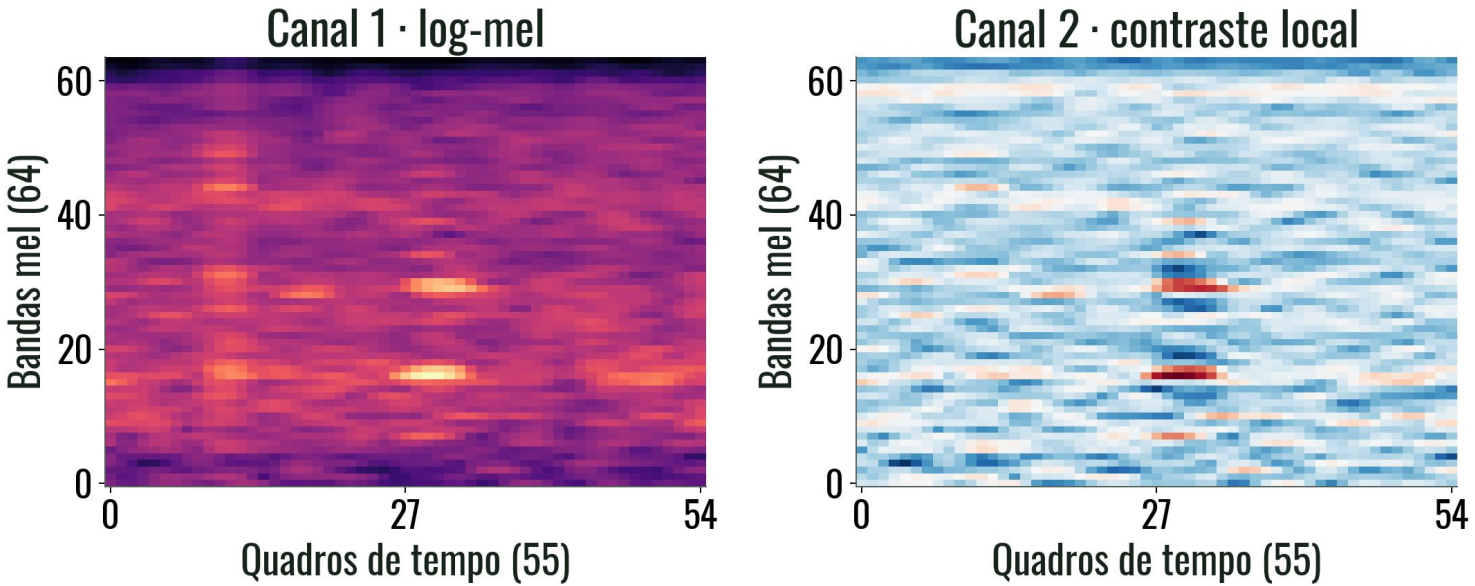
A STFT mostra frequência ao longo do tempo



FFT: 2.048 pontos · Hann: 128 ms · passo: 16 ms · espaçamento: 7,8125 Hz.

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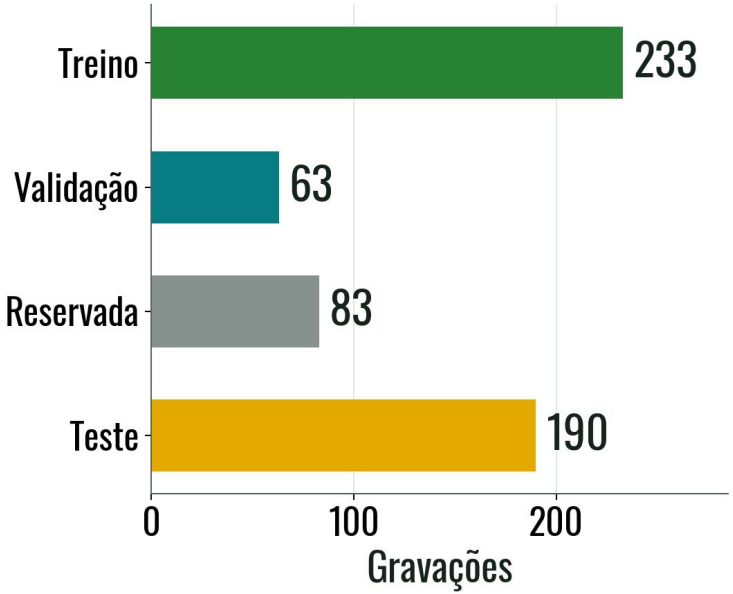
A entrada da rede tem dois canais

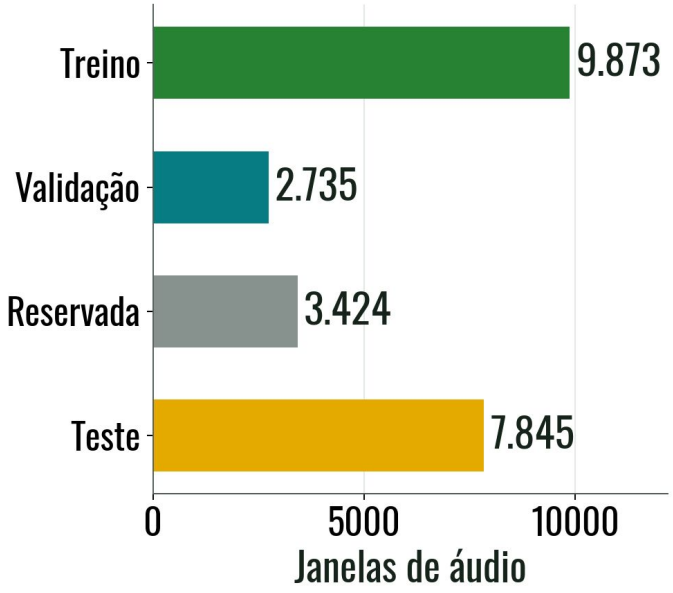


80–4.000 Hz · 128 filtros mel → 64 bandas · tensor final: 2 × 64 × 55.

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As partições são feitas por gravação



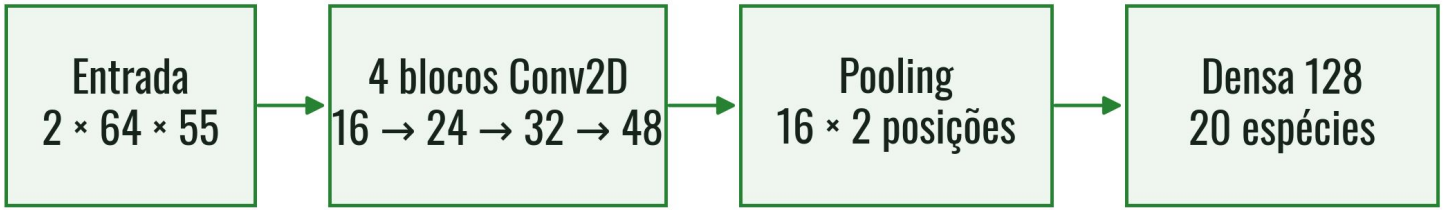


20 espécies · probabilidades das janelas são agregadas por gravação.

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A CNN aprende padrões no espectrograma

224.036 parâmetros treináveis

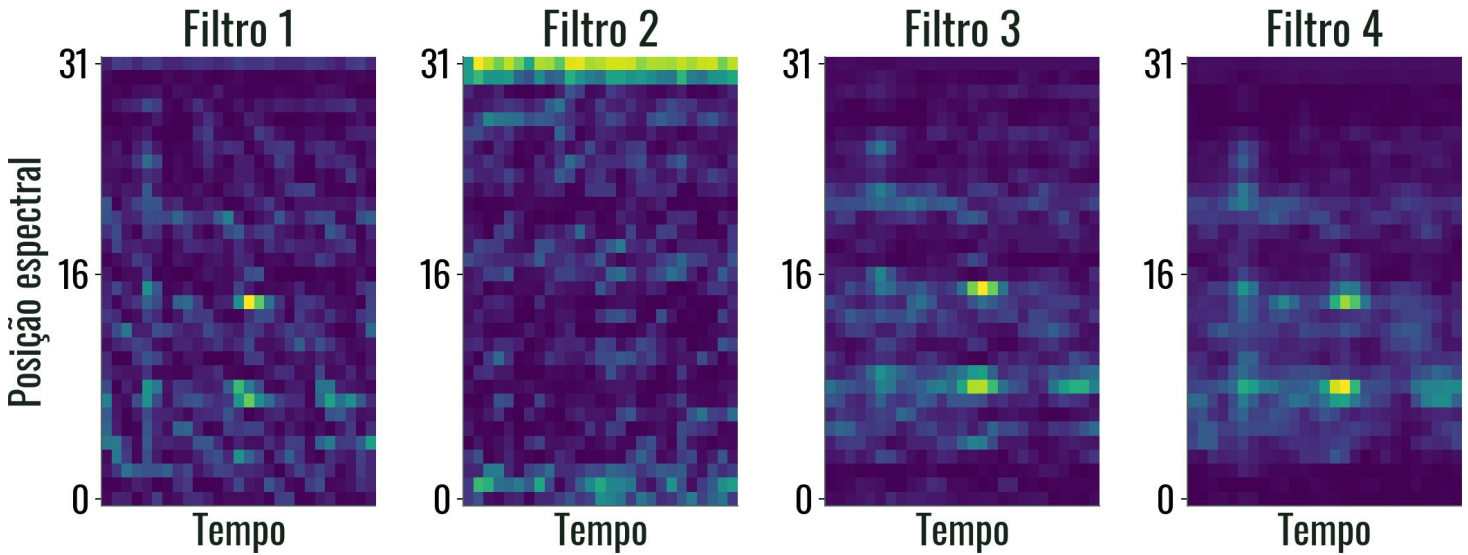


Conv 3 × 3 + BatchNorm + GELU + AvgPool

Convolução 3 × 3 · BatchNorm · GELU · pooling · saída para 20 espécies.

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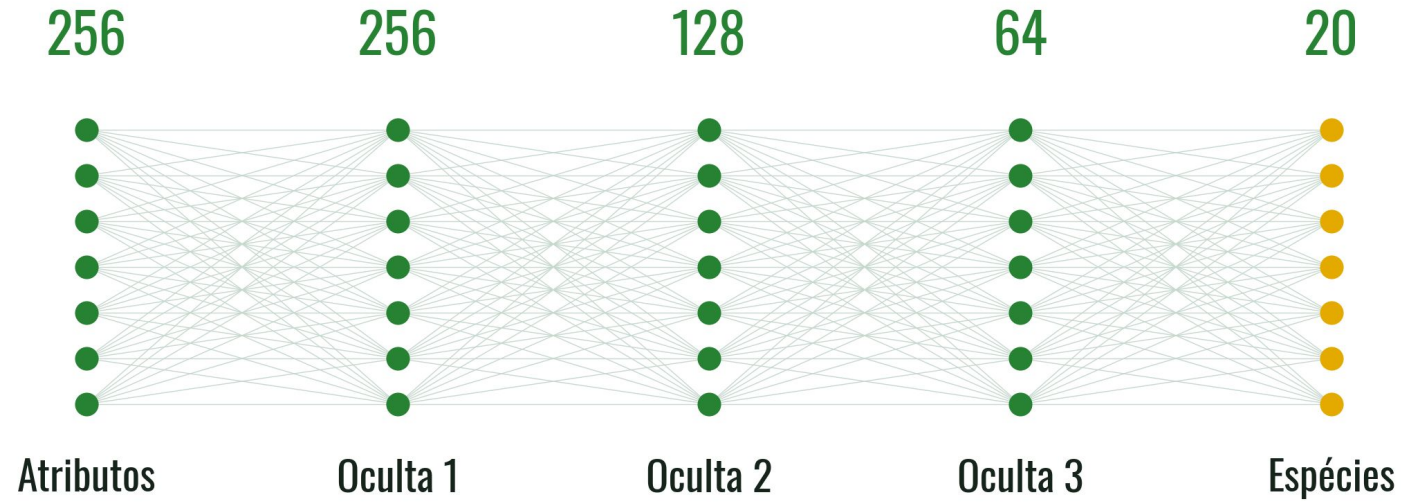
Os filtros produzem mapas de ativação



Quatro canais do primeiro bloco · pesos do checkpoint treinado · mesma janela real.

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A MLP aprende com atributos espectrais

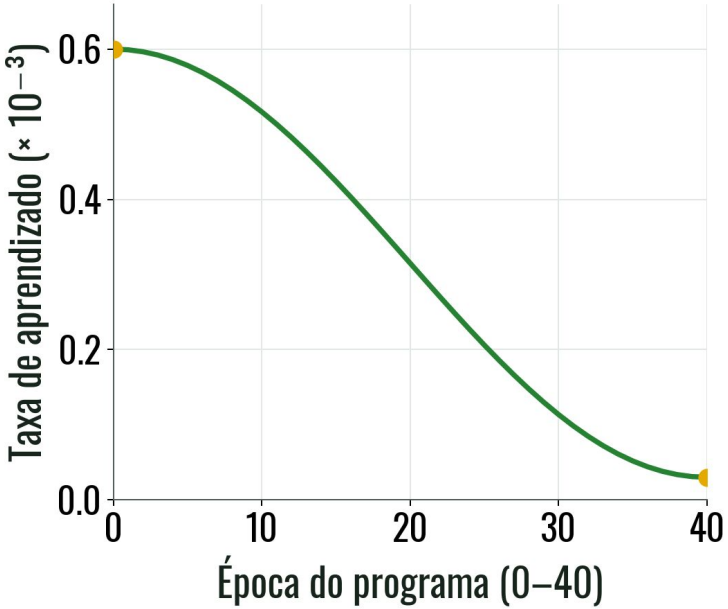


2 canais × 64 bandas × (média + desvio) = 256 atributos

256 atributos · camadas ocultas: 256 → 128 → 64 · 109.140 parâmetros.

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AdamW e cosseno controlam a atualização



AdamW
lr inicial = 0,0006

Decaimento de pesos
weight decay = 0,03

Gradientes estáveis
norma limitada a 3

Lote: 128 · lr: 0,0006 → 0,00003 · weight decay: 0,03 · clip de gradiente: 3.

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Regularização durante o treino

1 **Dropout nas redes**

CNN: 0,40 · MLP: 0,35 / 0,35 / 0,20.

2 **Suavização dos alvos**

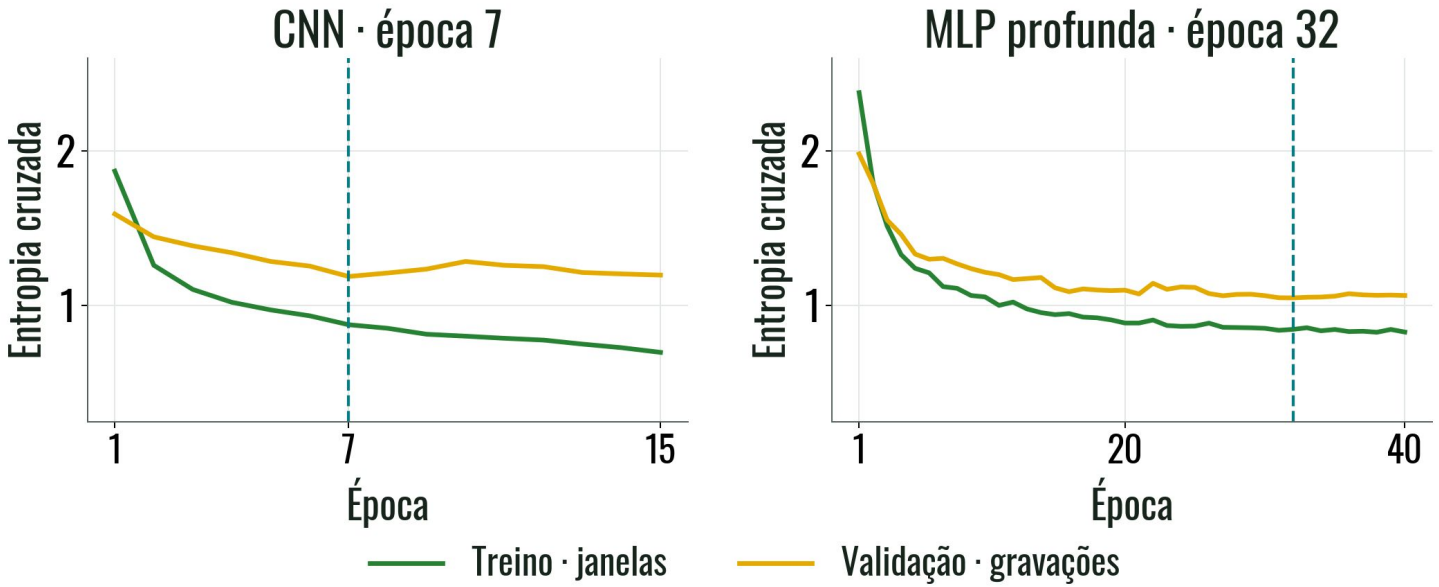
Label smoothing de 0,03 na entropia cruzada.

3 **Amostragem equilibrada**

Pesos equilibrados por espécie e por gravação.

Regularização nos neurônios, nos alvos e na escolha dos exemplos.

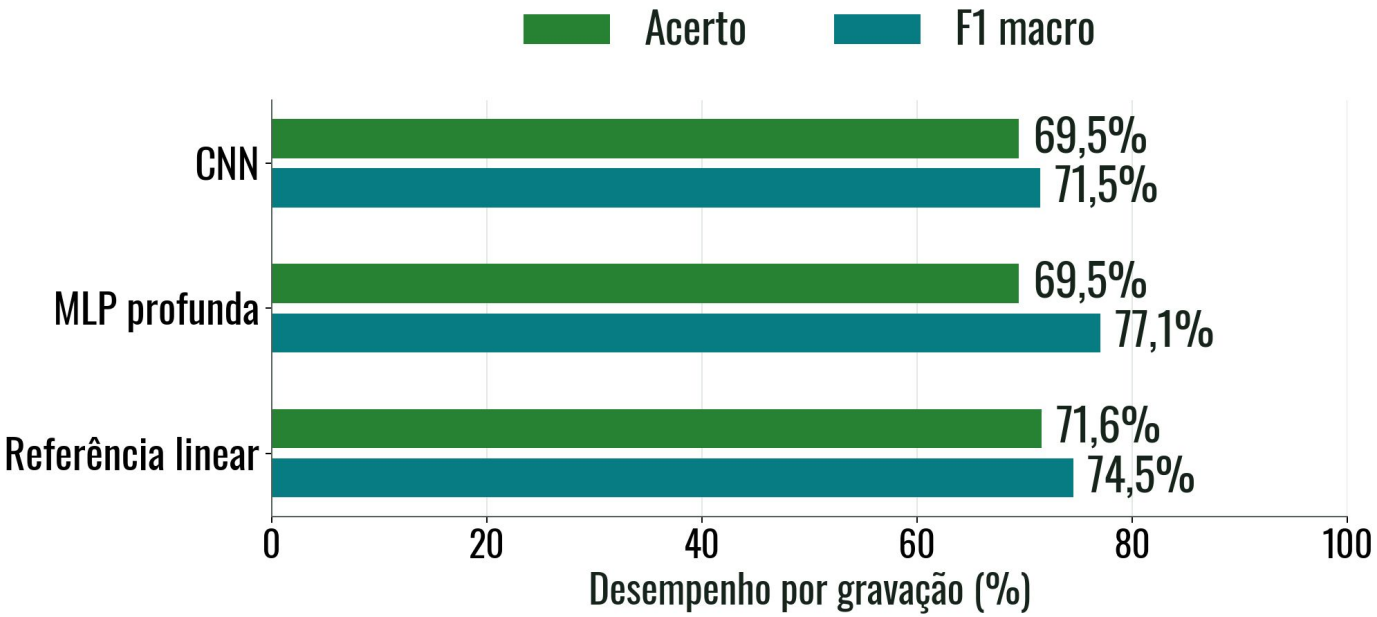
A validação escolhe o checkpoint



CNN: época 7 · MLP: época 32 · critério: menor perda por gravação na validação.

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Resultados por gravação: acerto e F1

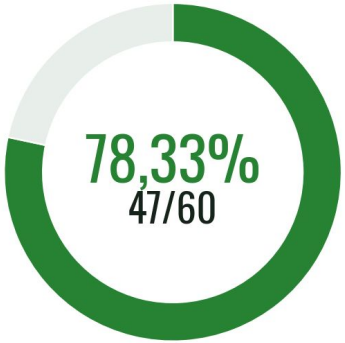


CNN e MLP: 132/190 acertos · referência linear: 136/190 · 20 espécies.

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Extensão: detector ternário por evento

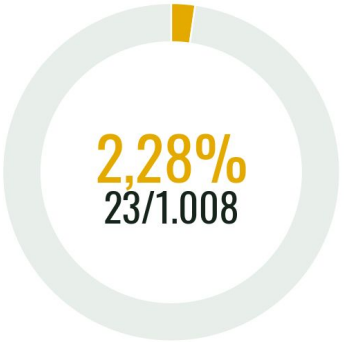
Aedes detectados



Ruído → Aedes



Outro mosquito → Aedes



Modelo clássico: HGB + regressão logística · 1.494 eventos · avaliação OOF por grupo.

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Referências e créditos

Mukundarajan et al. (2017) · áudio e celulares · eLife

Fanioudakis (2018) · redes profundas · EUSIPCO

Supratak et al. (2024) · MosquitoSong+ · PLOS ONE

Sinka et al. (2021) · HumBug e frequências publicadas

Dados, código e resultados · github.com/Lciarallo/mosquito-wingbeat

Foto do câmpus e modelo: IFSC · mosquito: CDC

Links completos, parâmetros e créditos nas notas de cada slide.

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Discussão final: limites e próximo ciclo

1 **Avaliação retrospectiva**

Validação com novas sessões e indivíduos pendente.

2 **Interpretação dos resultados**

Sem ablação para separar o efeito de cada técnica.

3 **Próximo ciclo: robustez**

Testar ruído, outros microfones e ambientes.

Na extensão ternária, o ruído sintético reduziu a detecção de Aedes.